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MIL-D-22233D
21 June 1991
SUPERSEDING
MIL-D-22233C
12 July 1977

MILITARY SPECIFICATION

DRILL RODS, COUPLING, SHANKS, ROCK BITS, AND EXTRACTOR, FOR DRIFTER DRILLS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers drill string components consisting of drill rods, couplings, shanks, replaceable rock bits, and extractor, for use with drifter drills.

1.2 Classification. The drill string components are one of the following types, styles, and sizes, as specified (see 6.2):

Type I - Drill rod.

Size A - 6-foot length.
Size B - 10-foot length.
Size C - 12-foot length.

Type II - Coupling.

Type III - Shank.

Style A - Tangential lug.
Style B - Four spline.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3820

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Type IV - Rock bit.

Style A - Cross design.
Style B - X-design.
Style C - Button design.

Size 2-1/2 - 2-1/2 inch gage.
Size 2-3/4 - 2-3/4 inch gage.
Size 3 - 3-inch gage.
Size 3-1/2 - 3-1/2 inch gage.
Size 4 - 4-inch gage.

Type V- Extractor.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116 - Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS39330 - Bit, Star, Rock Drill, X-Type, for Drifter Drills.
MS39335 - Bit, Button, Rock Drill; for Drifter Drills.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

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ASTM

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the ASTM, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC.

SAE J423 - Recommended Practice for Methods of Measuring Case Depth.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein. Material not definitely specified shall be the manufacturer's standard commercial quality. Steels shall have the necessary physical properties required for toughness, hardenability, and shock resistance. Components requiring carburization shall be thoroughly cleaned and dried to remove all oil, grease, lead oxide, paint, iron oxide, alkaline solutions, quenching salts, trichloroethylene residues, and other foreign substances before being carburized.

3.2 Standard commercial product. The drill string components shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the drill string components being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Type I - drill rod. Drill rod nominal overall length shall be 6 feet for size A, 10 feet for size B, and 12 feet for size C. Drill rod outside diameter (od) shall be not more than 1.502 inch and the air/water passageway shall be a nominal 0.500 inch in diameter. Threaded portion shall be 1-1/2 inch rope thread conforming to figure 3. Each end of the drill rod shall be threaded not less than 24 inches. Threads shall be turned at 90 degrees to centerline of the piece within 0.005 inch. Drill rod ends shall be square with thread within 0.005 inch. Drill rod corners shall be chamfered in accordance with good commercial practice. Full length of the rod shall be carburized inside and out.

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External surfaces of the rod shall be carburized to a case depth of not less than 0.045 inch but not more than 0.065 inch. Internal surfaces shall be carburized to a depth of not less than 0.035 inch. Rod shall have a case hardness between Rockwell hardness (Rh) 58C and Rh 65C. The manufacturer's standard means for maintaining thread lubrication shall be provided.

3.4 Type II - Coupling. Coupling dimensions shall be 2-3/16 inches ±1/16 inch in od by 7 inches ±1/16 inch in length. Internal threads conforming to figure 3 shall be mismatched at the coupling's center to stop rod movement through coupling. Mismatched threads shall not interfere with or prevent the ends of rods from butting against each other when coupled. Coupling shall have a case hardness between Rh 52C and Rh 58C.

3.5 Type III - shank. Shank shall be style A or B as specified herein. Style A shank shall conform to figure 1. Style B shank shall conform to figure 2. Shank material shall be alloy steel with the necessary properties required for toughness, hardness, hardenability, and shock resistance. All machined surfaces shall be finished to 125 microns. Figure dimensions are in inches. Unless otherwise specified, tolerances shall be: Fractional ±1/32 inch, decimal ±0.005, angular ±1 degree. The shank ends shall be square with the thread within 0.005 inch, the thread shall be turned at 90 degrees to centerline of piece, and all diameters shall be concentric within 0.005 inch. The shank end to be coupled to the drill steel shall be threaded with 1-1/2-inch rope thread conforming to figure 3. External and internal surfaces shall be carburized to a case depth of not less than 0.040 inch but not more than 0.080 inch. Shanks shall have a case hardness between Rh 56C and Rh 63C. The manufacturer's standard provisions for retaining the flushing tube seal shall be provided for the style B shank.

3.6 Type IV - rock bit. All bit bodies shall be of alloy steel having a case hardness of not less than Rh 42C. The face of style C bits shall have a case hardness of not less than Rh 36C. Brazing material used to secure the tungsten carbide inserts to the body of the bits shall be 40 to 50 percent silver-bearing alloy or not less than 40 percent copper alloy. Bonding characteristics of the copper alloy shall be not less than those of the silver-bearing alloy.

3.6.1 Style A bit. Style A bit shall conform to MS 39330, except the tungsten carbide inserts shall be positioned on the face of the bit to form a cross configuration.

3.6.2 Style B bit. Style B bit shall conform to MS39330.

3.6.3 Style C bit. Style C bit shall conform to MS39335. The bit shall be capable of drilling through 600 feet of Barre granite before destruction.

3.7 Type V - extractor. Extractor shall have the capability to score and grip broken carburized drill rods and couplings for the purpose of withdrawing a broken drill string from a hole. Internal threads, at the drill rod end, shall conform to figure 3. Internal threads, at the extractor end, shall be of the manufacturer's design.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

* 4.2 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one each of the type, style, and size specified. All components of similar classification offered for delivery at one time shall be considered a lot for the purpose of inspection.

* 4.2.1 Sampling for examination. Recommended inspection level and an Acceptable Quality Level (AQL) is provided in 6.5.1.

* 4.2.2 Sampling for tests. Recommended inspection level and an AQL is provided in 6.5.2.

4.3 Examination. Each sample selected in accordance with 4.2.1 shall be examined for compliance with the requirements in section 3 of this specification. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement shall constitute one defect.

4.4 Tests. Samples selected in accordance with 4.2.2 shall be gaged for size, tested for core and case hardness, and the case depth of carburization shall be measured in accordance with SAE J423 to determine the degree of compliance with the applicable portions of this specification. Failure to meet specified requirements shall constitute one defect for each failed requirement.

4.5 Packaging inspection. The preservation, packing, and marking of the components shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. All unpainted exterior surfaces, including threaded surfaces, of the drill rods, couplings, shanks, extractors, and bits shall be coated with type P-1 preservative conforming to the applicable specification listed in, and applied in accordance with MIL-P-116.

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* 5.1.2 Commercial. Material shall be packaged in accordance with ASTM D 3951.

* 5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A. The drill rods, couplings, shanks, extractors and bits shall be packed in boxes conforming to PPP-B-621, class 2. Items of like description shall be packed together. The boxes for drill rods shall be modified to include full double ends, with the inner pieces positioned vertically and the outer pieces horizontally. The minimum thickness of the end pieces shall be governed by the minimum thickness of ends specified for styles 2 or 2-1/2 boxes of the above specification. The contents shall be cushioned, blocked, and braced to prevent movement within the boxes.

5.2.2 Level B. The drill rods, couplings, shanks, extractors, and bits shall be packed as specified for level A, except that the boxes shall conform to PPP-B-621, class 1.

* 5.2.3 Commercial. Material shall be packed in accordance with ASTM D 3951.

* 5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

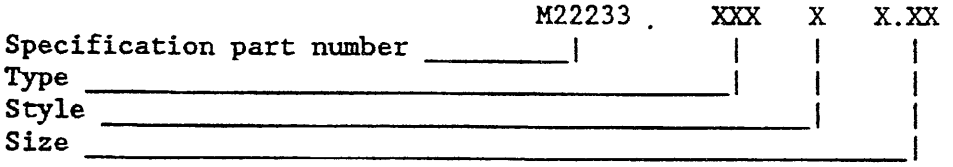
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components covered by this specification, drill rod, coupling, shank, extractor, and replaceable rock bit are for equipping drifter drills to drill holes in rock formations.

6.2 Acquisition documents. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, styles, and size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Level of preservation and level of packing required (see 5.1 and 5.2).

* 6.3 Part or identifying number (PIN). The PIN to be used for drill string components covered by this specification are created as follows:



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6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) invokes the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

* 6.5 Sampling procedures.

* 6.5.1 Sampling for examination. Recommended inspection level is II and an AQL of 4.0 percent defective (see 4.2.1).

* 6.5.2 Sampling for tests. Recommended inspection level is S-4 and an AQL of 4.0 percent defective (see 4.2.2).

* 6.6 Cross-reference of classification. This revision contains no changes in the classification of the drill string components. The following data is a historical record of classification changes made in previous revisions of this specification:

MIL-D-22233B	MIL-D-22233C
Type I - Drill rod.	Type I- Drill rods. Size A - 6-foot length. Size B - 10-foot length. Size C - 12-foot length.

* 6.7 Subject term (key word) listing.

Drill string
Carburized
Barre granite

6.8 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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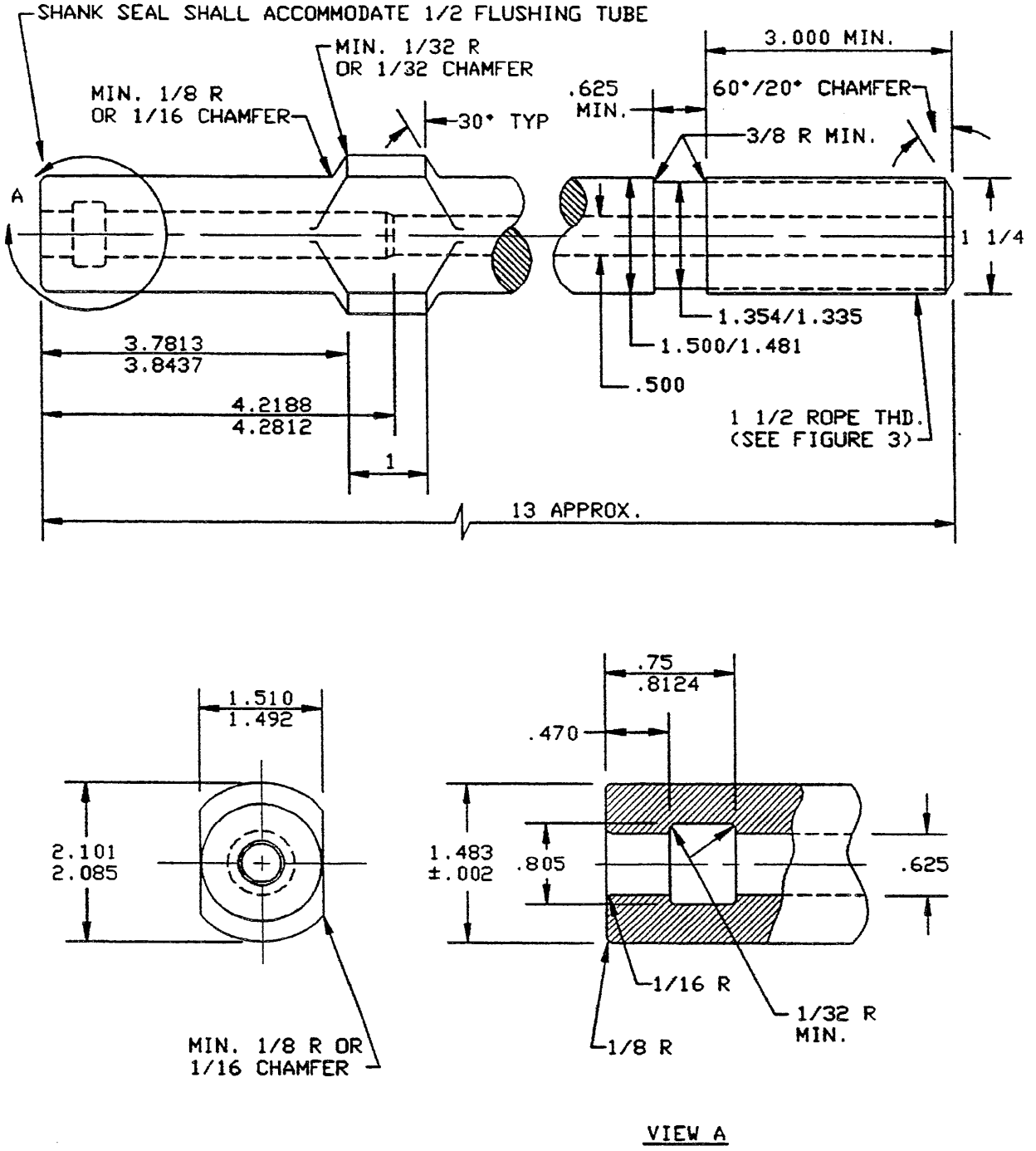


FIGURE 1. 1-1/2 inch tangential lug shank.

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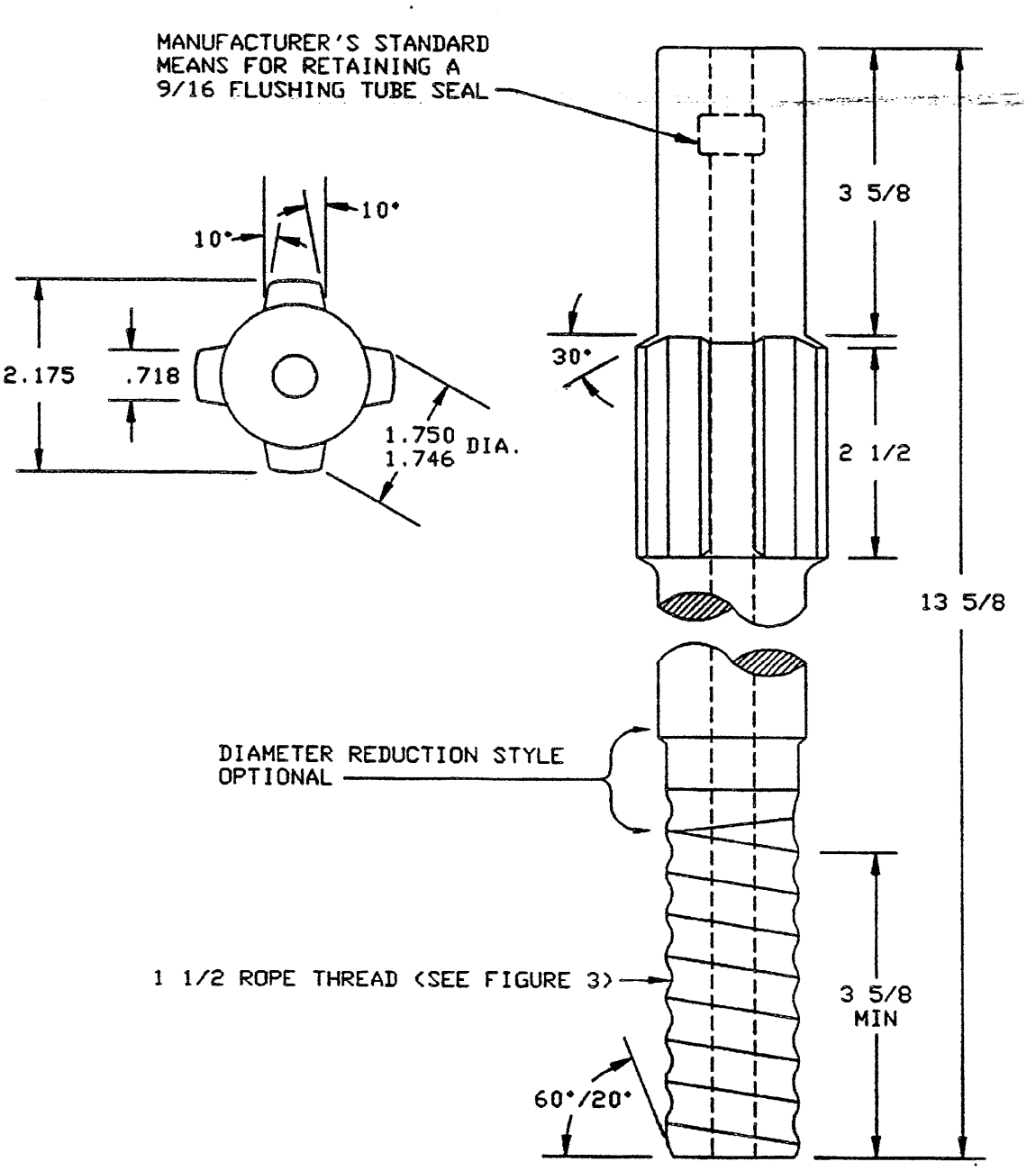
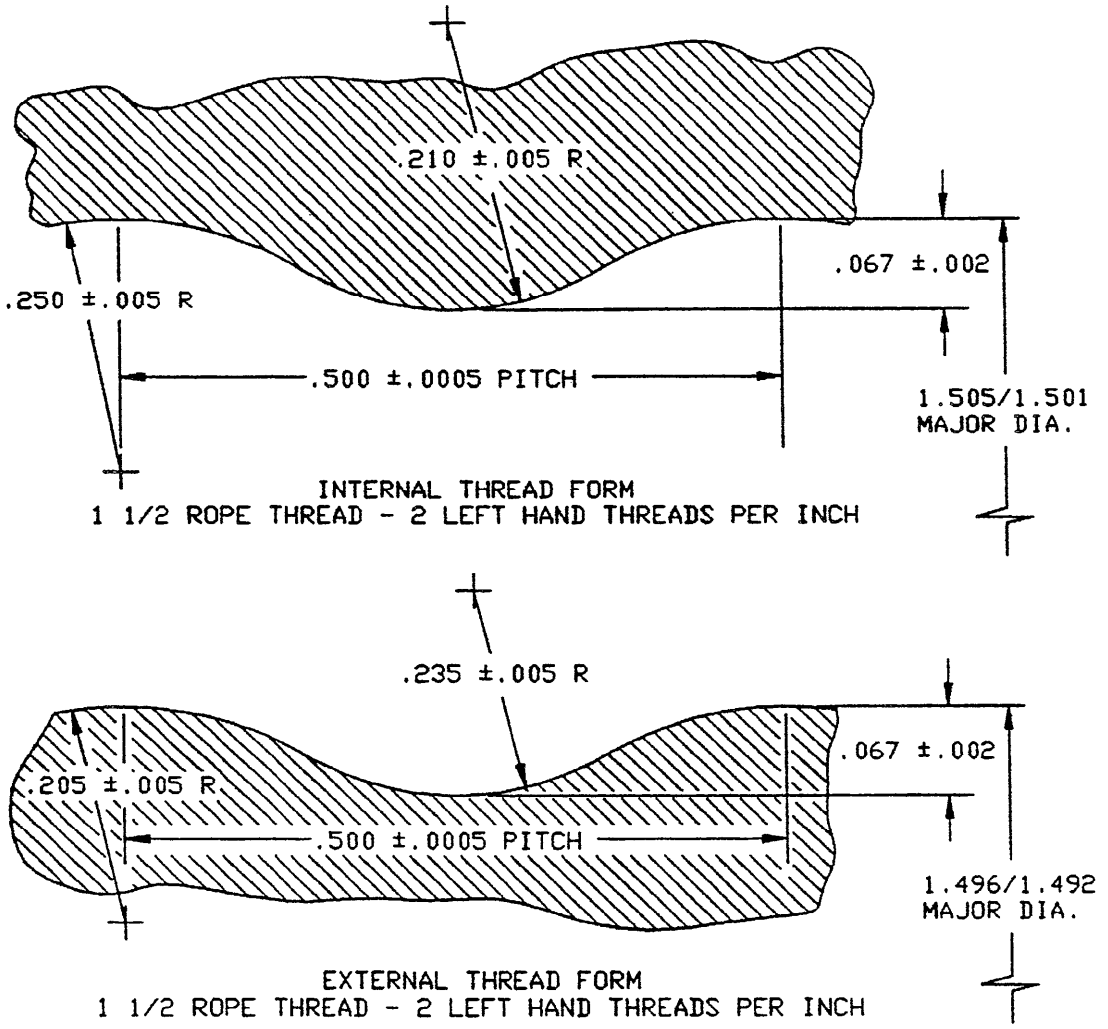


FIGURE 2. 1 3/4 inch four spline shank.

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Notes:

1. The thread forms shown are considered ideal thread forms. Suppliers of drill rods, couplings, bits, and shanks need not supply thread forms conforming to the exact dimension shown. However, drill rods, couplings, bits and shanks must be supplied with a thread form capable of being properly mated to, used, and removed from the applicable thread form, as shown, by hand with no more assistance than that afforded by the used of a 12-inch pipe wrench.
2. All machined surfaces shall be finished to 125 microns.
3. Ends must be square with thread within .005 inch and thread must be turned at 90° to centerline of piece within .005 inch.
4. All dimensions are in inches. Unless otherwise specified, tolerances shall be: fractional ±1/32 inch, decimal ±.005 inch, and angular ±1°.

FIGURE 3. Rope type thread form for drill rods, couplings, bits, and shanks.

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INSTRUCTIONS

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-D-22233D	910621

3. DOCUMENT TITLE

Drill Rods, Coupling, Shanks, Rock Bits, and Extractor for Drifter Drills

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Commmanding Officer	(805) 982-5926	551-5926
c. ADDRESS (Include Zip Code) NCBC (1562A) Port Hueneme, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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